

ALLAN ILYASOV

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SUMMARY

Graduate student in Data Science at St. John's University with hands-on experience building and deploying AI-powered applications. Skilled in machine learning, full-stack development, and cloud infrastructure, with a track record of taking projects from idea to production. Passionate about using software and AI to tackle problems that benefit society, from making education more accessible to improving how people interact with technology.

EXPERIENCE

Graduate Research Assistant – St. John's University

September 2025 – Present

Codify AI – Advancing Education Through Generative AI & LLM (Dr. Fazel Keshtkar):

- Product manager & lead architect of codifyai.org, an AI-powered programming tutor using LangChain agents, Claude Haiku 4.5 via AWS Bedrock, and RAG for streaming responses, and context-aware tutoring in Java, Python & SQL.
- Designed production-ready AWS infrastructure (Elastic Beanstalk, DynamoDB, Route 53, HTTPS) with Docker containerization, supporting scalable REST API architecture.
- Built React-based adaptive learning platform with persistent state management, real-time progress tracking.
- Earned “Commercializable Project” award at the 2025 St. John's University's Student Research Conference.
- Presented at the Middle States Commission on Higher Education (MSCHE) Conference. stjohns.edu/news-media

Manager – Uncle Edik's Pickles

December 2019 – Present

- Grew a home-based pickle start-up and supported its growth into a brick-and-mortar factory and national brand.
- Trained & led a team of 5 employees in production of thousands of jars per week of 9 different pickled products.
- Managed sales to customers, wholesale delivery to supermarkets, and inventory replenishment from Restaurant Depot.
- Designed branded merchandise, and produced viral marketing content, generating millions of views.
- Ensure compliance with NYS Food Safety regulations & maintained quality control with daily production checklists.

SELECT PROJECTS

Time Series Forecasting for Financial Markets | github.com/allanilya/Stock-analytics

- Created ARIMA, LSTM, & GRU forecasting models across AAPL, NVDA, & LYFT, reducing RMSE by up to 82%.
- Implemented grid search across 36 ARIMA configurations and built 2-layer LSTM/GRU architectures with dropout regularization utilizing Python, TensorFlow, and Statsmodels.

Multimodal Video Analysis | github.com/allanilya/Multimodal-Video-Analysis

- Developed full-stack AI video analysis platform with Flask/Next.js featuring CLIP-based visual search, transcript search, and Gemini 2.5 Flash multimodal chat with conversational memory.
- Engineered async parallel processing pipeline achieving <3s response time via background frame extraction.

Diabetes Prediction Model | github.com/allanilya/Diabetes-Prediction

- Developed Random Forest, Logistic Regression & KNN models with 5-fold cross-validation to predict diabetes risk.
- Investigated lifestyle and demographic predictors of diabetes using 230K+ patient records.
- Prepared data through imputation, encoding, outlier detection and addressed class imbalance leveraging SMOTE.

EDUCATION

St. John's University – Queens, New York

Master of Science, Data Science

May 2026

Bachelor of Science, Computer Science

May 2025

- Dean's List 2021-2025 | Member of the Association for Computing Machinery | EPIC Mentoring Program, Student Mentor

SKILLS SUMMARY

- **Programming Languages:** Python, Java, SQL, R, JavaScript, Swift, TypeScript, C++, PHP
- **Frameworks & Libraries:** React, Next.js, Django, Flask, Spring Boot, Node.js, Bootstrap, LangChain, LangGraph, REST APIs, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch, Spark, Beautiful Soup, NLTK, Spacy
- **Cloud & DevOps:** AWS, Docker, DynamoDB, Elastic Beanstalk, Lambda, API Gateway
- **Linguistic Skills:** Russian native speaker | Intermediate proficiency in Spanish

Work Eligibility: US Citizen